

Database Systems (CS2005)

Week – 08

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CHAPTER 3

Data Modeling Using the Entity-Relationship (ER) Model

Overview of Database Design Process

- Two main activities
 - Database design
 - Applications design
- Focus in this chapter on conceptual database design
 - To design the conceptual schema for a database application
- Applications design focuses on the programs and interfaces that access the database
 - Generally considered part of software engineering

Overview of Database Design Process

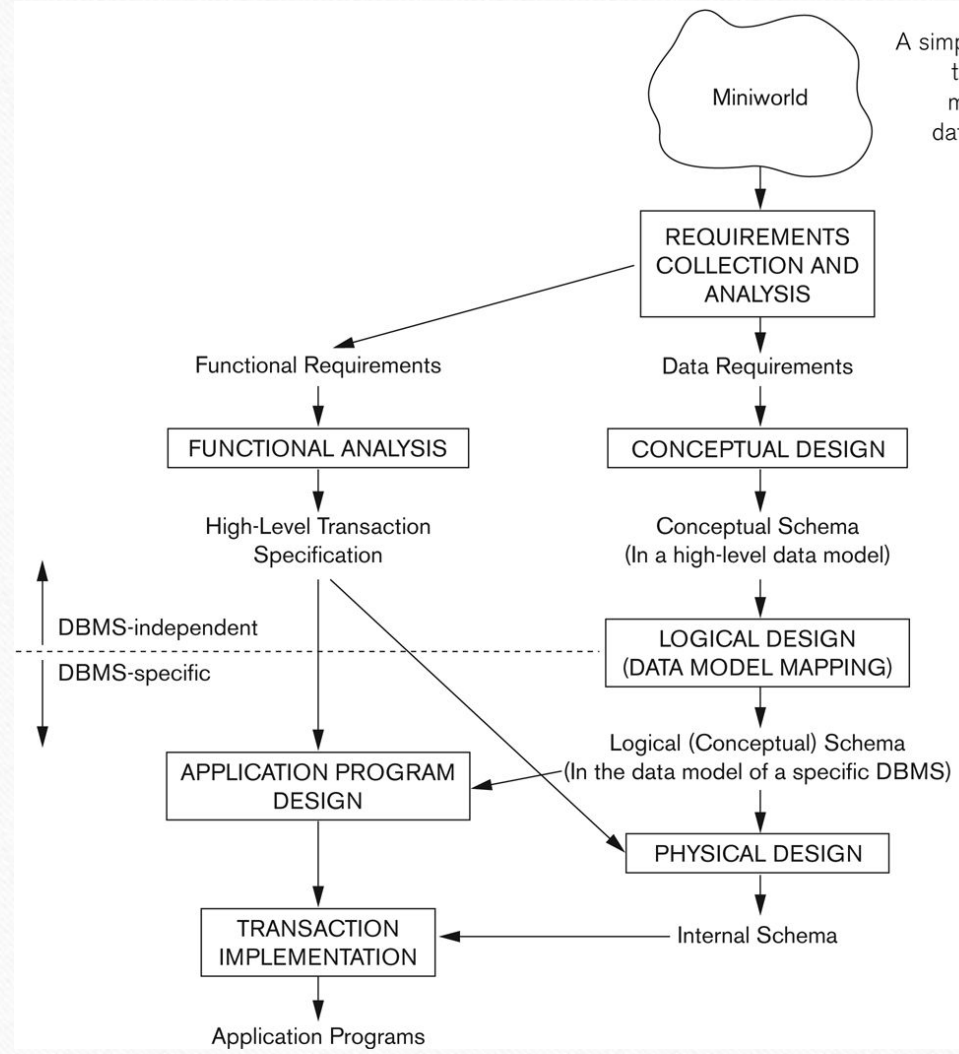


Figure 3.1
A simplified diagram
to illustrate the
main phases of
database design.

Methodologies for Conceptual Design

- Entity Relationship (ER) Diagrams (This Chapter)
- Enhanced Entity Relationship (EER) Diagrams (Chapter 4)
- Use of Design Tools in industry for designing and documenting large scale designs
- The UML (Unified Modeling Language) Class Diagrams are popular in industry to document conceptual database designs

Example COMPANY Database

- We need to create a database schema design based on the following (simplified) **requirements** of the COMPANY Database:
 - The company is organized into DEPARTMENTS. Each department has a name, number and an employee who *manages* the department. We keep track of the start date of the department manager. A department may have several locations.
 - Each department *controls* a number of PROJECTs. Each project has a unique name, unique number and is located at a single location.

Example COMPANY Database (Continued)

- The database will store each EMPLOYEE's social security number, address, salary, gender, and birthdate.
 - Each employee *works for* one department but may *work on* several projects.
 - The DB will keep track of the number of hours per week that an employee currently works on each project.
 - It is required to keep track of the *direct supervisor* of each employee.
- Each employee may *have* a number of DEPENDENTS.
 - For each dependent, the DB keeps a record of name, gender, birthdate, and relationship to the employee.

ER Model Concepts

- Entities and Attributes
 - Entity is a basic concept for the ER model. Entities are specific things or objects in the mini-world that are represented in the database.
 - For example the EMPLOYEE John Smith, the Research DEPARTMENT, the ProductX PROJECT
 - Attributes are properties used to describe an entity.
 - For example an EMPLOYEE entity may have the attributes Name, SSN, Address, gender, BirthDate
 - A specific entity will have a value for each of its attributes.
 - For example a specific employee entity may have Name='John Smith', SSN='123456789', Address
='731, Fondren, Houston, TX', gender='M', BirthDate='09 JAN 55'
 - Each attribute has a *value set* (or data type) associated with it – e.g. integer, string, date, enumerated type, ...

Types of Attributes (1)

- Simple
 - Each entity has a single atomic value for the attribute. For example, SSN or gender.
- Composite
 - The attribute may be composed of several components. For example:
 - Address(Apt#, House#, Street, City, State, ZipCode, Country), or
 - Name(FirstName, MiddleName, LastName).
 - Composition may form a hierarchy where some components are themselves composite.

- Multi-valued

Types of Attributes (2)

- In general, composite attributes may be nested arbitrarily to any number of levels, although this is rare.
 - For example, PreviousDegrees of a STUDENT is a composite multi-valued attribute denoted by {PreviousDegrees (College, Year, Degree, Field)}
 - Multiple PreviousDegrees values can exist
 - Each has four subcomponent attributes:
 - College, Year, Degree, Field

Example of a composite attribute

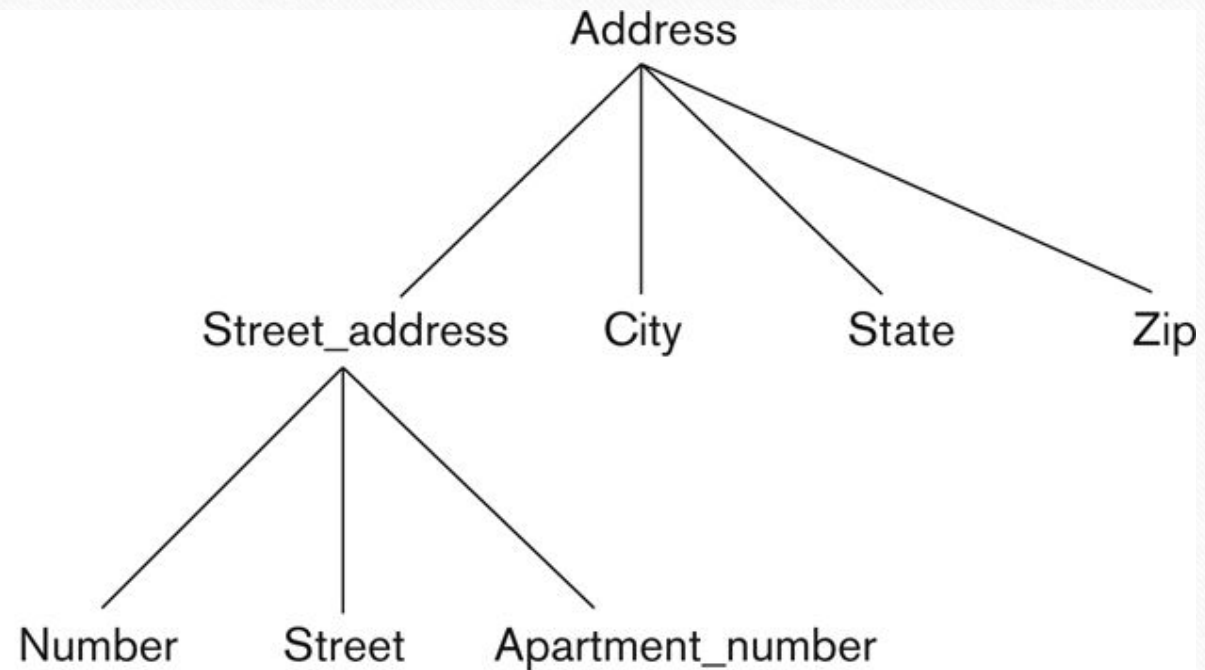


Figure 3.4

A hierarchy of composite attributes.

Entity Types and Key Attributes (1)

- Entities with the same basic attributes are grouped or typed into an entity type.
 - For example, the entity type EMPLOYEE and PROJECT.
- An attribute of an entity type for which each entity must have a unique value is called a key attribute of the entity type.
 - For example, SSN of EMPLOYEE.

Entity Types and Key Attributes (2)

- A key attribute may be composite.
 - VehicleTagNumber is a key of the CAR entity type with components (Number, State).
- An entity type may have more than one key.
 - The CAR entity type may have two keys:
 - VehicleIdentificationNumber (popularly called VIN)
 - VehicleTagNumber (Number, State), aka license plate number.

- Each key is underlined (Note: this is different from the relational schema where only one “primary key is

Entity Set

- Each entity type will have a collection of entities stored in the database
 - Called the **entity set** or sometimes **entity collection**
- Same name (CAR) used to refer to both the entity type and the entity set
- However, entity type and entity set may be given different names
- Entity set is the current *state* of the entities of that type that are stored in the database

Value Sets (Domains) of Attributes

- Each simple attribute is associated with a value set
 - E.g., Lastname has a value which is a character string of upto 15 characters, say
 - Date has a value consisting of MM-DD-YYYY where each letter is an integer
- A **value set** specifies the set of values associated with an attribute

Attributes and Value Sets

- Value sets are similar to data types in most programming languages – e.g., integer, character (n), real, bit
- Mathematically, an attribute A for an entity type E whose value set is V is defined as a function

$$A : E \rightarrow P(V)$$

Where $P(V)$ indicates a power set (which means all possible subsets) of V . The above definition covers simple and multivalued attributes.

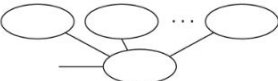
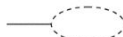
- We refer to the value of attribute A for entity e as $A(e)$.

Displaying an Entity type

- In ER diagrams, an entity type is displayed in a rectangular box
- Attributes are displayed in ovals
 - Each attribute is connected to its entity type
 - Components of a composite attribute are connected to the oval representing the composite attribute
 - Each key attribute is underlined
 - Multivalued attributes displayed in double ovals

NOTATION for ER diagrams

Figure 3.14
Summary of the
notation for ER
diagrams.

Symbol	Meaning
	Entity
	Weak Entity
	Relationship
	Identifying Relationship
	Attribute
	Key Attribute
	Multivalued Attribute
	Composite Attribute
	Derived Attribute
	Total Participation of E_2 in R
	Cardinality Ratio 1: N for $E_1:E_2$ in R
	Structural Constraint (min, max) on Participation of E in R

Entity Type CAR with two keys and a corresponding Entity Set

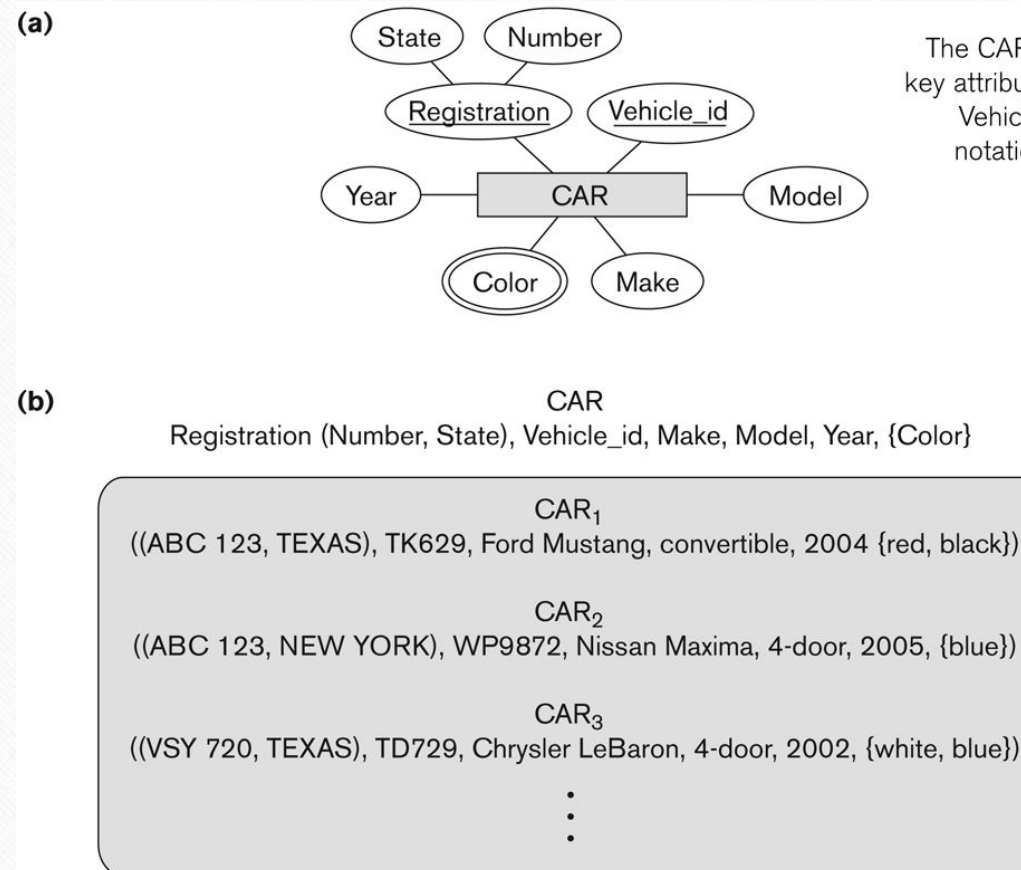


Figure 3.7

The CAR entity type with two key attributes, Registration and Vehicle_id. (a) ER diagram notation. (b) Entity set with three entities.

Single Line & Double Line in ERD

- A single line in an ERD represents a **partial participation** or an **optional relationship** between entities. This means that an entity is not required to participate in the relationship, and its participation is optional.
- A double line in an ERD represents **total participation** or a **mandatory relationship**. This means that every instance of the entity must participate in the relationship, and it cannot exist without being involved in that relationship.

Single Line & Double Line in ERD

- Single: Person — owns — Car
- Double: Order == contains == Order Item

Weak entity

- A **weak entity** is an entity that cannot exist independently in a database without being associated with another (strong) entity
- **Customer** and **Order**

Identifying Relationship

- An **identifying relationship** is a relationship between a strong entity and a weak entity in which the weak entity depends on the strong entity for its existence.
- **Customer and Order**

Initial Conceptual Design of Entity Types for the COMPANY Database Schema

- Based on the requirements, we can identify four initial entity types in the COMPANY database:
 - DEPARTMENT
 - PROJECT
 - EMPLOYEE
 - DEPENDENT
- The initial attributes shown are derived from the requirements description

Initial Design of Entity Types: EMPLOYEE, DEPARTMENT, PROJECT, DEPENDENT

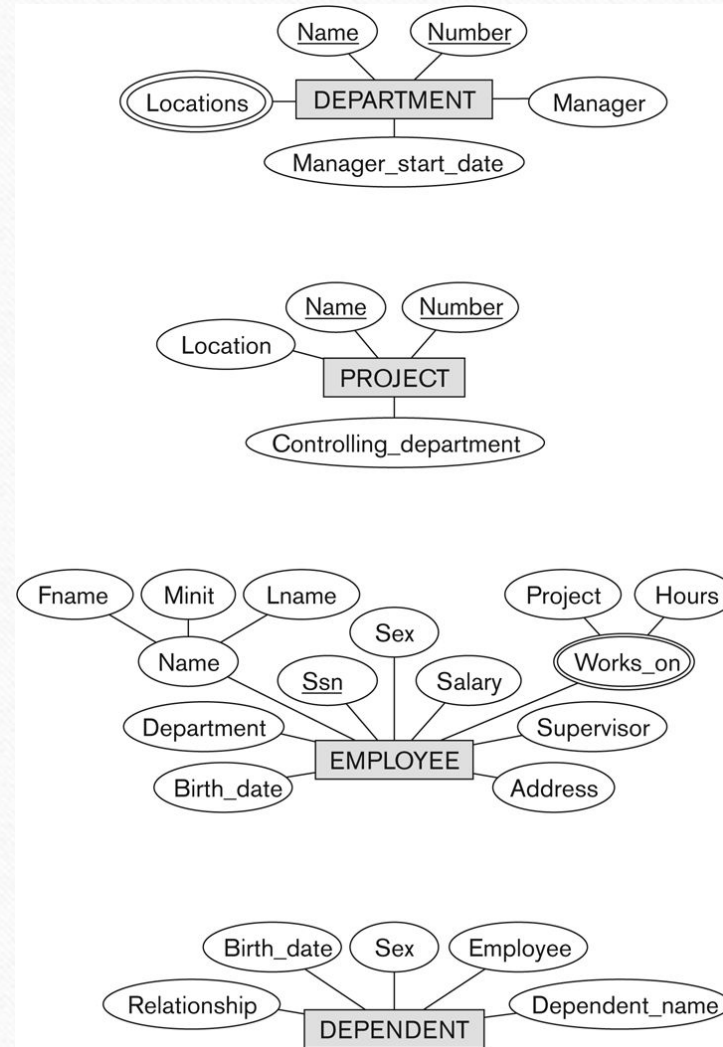


Figure 3.8
Preliminary design of entity
types for the COMPANY
database. Some of the
shown attributes will be
refined into relationships.

Refining the initial design by introducing **relationships**

- The initial design is typically not complete
- Some aspects in the requirements will be represented as **relationships**
- ER model has three main concepts:
 - Entities (and their entity types and entity sets)
 - Attributes (simple, composite, multivalued)
 - Relationships (and their relationship types and relationship sets)

Relationships and Relationship Types (1)

- A **relationship** relates two or more distinct entities with a specific meaning.
 - For example, EMPLOYEE John Smith *works on* the ProductX PROJECT, or EMPLOYEE Franklin Wong *manages* the Research DEPARTMENT.
- Relationships of the same type are grouped or typed into a **relationship type**.
 - For example, the WORKS_ON relationship type in which EMPLOYEEs and PROJECTs participate, or the MANAGES relationship type in which EMPLOYEEs and DEPARTMENTs participate.
- The degree of a relationship type is the number of participating entity types.
 - Both MANAGES and WORKS_ON are *binary* relationships.

One-To-One Relationship

- A *one-to-one* (1:1) relationship means that each record in Table A relates to one, and only one, record in Table B, and each record in Table B relates to one, and only one, record in Table A.

PERSONAL						
EmployeeID	FirstName	LastName	Address	City	State	Zip
EN1-10	Carol	Schaaf	2306 Palisade Ave.	Union City	NJ	07087
EN1-12	Gayle	Murray	1855 Broadway	New York	NY	12390
EN1-15	Steve	Baranco	742 Forrest St.	Kearny	NJ	07032
EN1-16	Kristine	Racich	416 Bloomfield St.	Hoboken	NJ	07030
EN1-19	Barbara	Zumbo	24 Central Ave.	Ritchfield Park	NJ	07660
EN1-20	Daniel	Gordon	2 Angelique St.	Weehawken	NJ	07087
EN1-22	Jacqueline	Rivet	3600 Bergeline Ave.	Union City	NJ	07087
EN1-23	Betsy	Roslyn	1800 Boulevard East	Weehawken	NJ	07086
EN1-25	Will	Strick	2100 91st St.	North Bergen	NJ	07047
EN1-26	Susan	Shipe	240 Fifth Ave.	New York	NY	10018

PAYROLL	
EmployeeID	PayRate
EN1-10	\$25.00
EN1-12	\$27.50
EN1-15	\$20.00
EN1-16	\$19.00
EN1-19	\$22.75
EN1-20	\$23.00
EN1-22	\$22.50
EN1-23	\$19.50
EN1-25	\$12.50
EN1-26	\$14.00

One-To-Many Relationship

- A *one-to-many* (1:N) relationship means a record in Table A can relate to zero, one, or many records in Table B. Many records in Table B can relate to one record in Table A. The *potential* relationship is what's important; for a single record in Table A, there might be no related records in Table B, or there might be only one related record, but there *could* be many.

CUSTOMERS

CustomerID	CustomerName	Address	City	State	Zip
20151	Engel's Books	19 International Dr	Ryebrook	NY	10273-9764
20493	Jamison Books	396 Apache Ave	Fountain Valley	CA	92708-4982
20512	Gardening Galore	79 Gessner Pk	Houston	TX	77024-6261
20688	Books Abound	51 Ulster St	Denver	CO	80237-3386
20784	Book World	687 Mountain Rd	Stowe	VT	08276-3196
20926	The Corner Booksotre	36 N.Miller Ave	Syracuse	NY	13206-4976
20932	Allendale Books	512 Columbia Rd	Someville	NJ	08876-2987
21570	In Between the Covers	2008 Delta Ave	Cincinnati	OH	45208-4468
21587	Books and Beyond	51 Windsor St	Cambridge	MA	02139-2123
21965	Cover to Cover	12 Harbor St	Burlington	VT	04982-2977

ORDERS

OrderNum	CustomerID	OrderDate	ShipDate	Shipper
76654	20151	2/1/00	2/6/00	USPS
74432	20151	6/30/99	7/2/99	Federal Express
75987	20151	11/10/99	11/12/99	UPS
62922	20493	9/5/99	9/6/99	UPS
65745	20493	10/1/99	10/3/99	USPS
72212	20493	4/22/00	4/25/00	UPS
73547	20493	8/17/99	8/20/99	UPS
69211	21570	5/12/99	5/12/99	Federal Express
70343	21587	10/2/00	10/4/00	UPS
72833	21587	12/14/99	12/17/99	UPS

Many-To-Many Relationship

EMPLOYEES

EmployeeID	Last Name	First Name	ProjectNum
EN1-26	O'Brien	Sean	30-452-T3
EN1-26	O'Brien	Sean	30-457-T3
EN1-26	O'Brien	Sean	31-124-T3
EN1-33	Guya	Amy	30-452-T3
EN1-33	Guya	Amy	30-482-TC
EN1-33	Guya	Amy	31-124-T3
EN1-35	Baranco	Steven	30-452-T3
EN1-35	Baranco	Steven	31-238-TC
EN1-36	Roslyn	Elizabeth	35-152-TC
EN1-38	Schaaf	Carol	36-272-TC
EN1-40	Wing	Alexandra	31-238-TC
EN1-40	Wing	Alexandra	31-241-TC

PROJECTS

ProjectNum	ProjectTitle	EmployeeID
30-452-T3	Woodworking Around The House	EN1-26
30-452-T3	Woodworking Around The House	EN1-33
30-452-T3	Woodworking Around The House	EN1-35
30-457-T3	Basic Home Electronics	EN1-26
30-482-TC	The Complete American Auto Repair Guide	EN1-33
31-124-T3	The Sport Of Hang Gliding	EN1-26
31-124-T3	The Sport Of Hang Gliding	EN1-33
31-238-TC	The Complete Baseball Reference	EN1-35
31-238-TC	The Complete Baseball Reference	EN1-35
31-241-TC	Improving Your Tennis Game	EN1-40
35-152-TC	Managing Your Personal Finances	EN1-36
36-272-TC	Using Electronic Mail Effectively	EN1-38

Relationship instances of the WORKS_FOR N:1 relationship between EMPLOYEE and DEPARTMENT

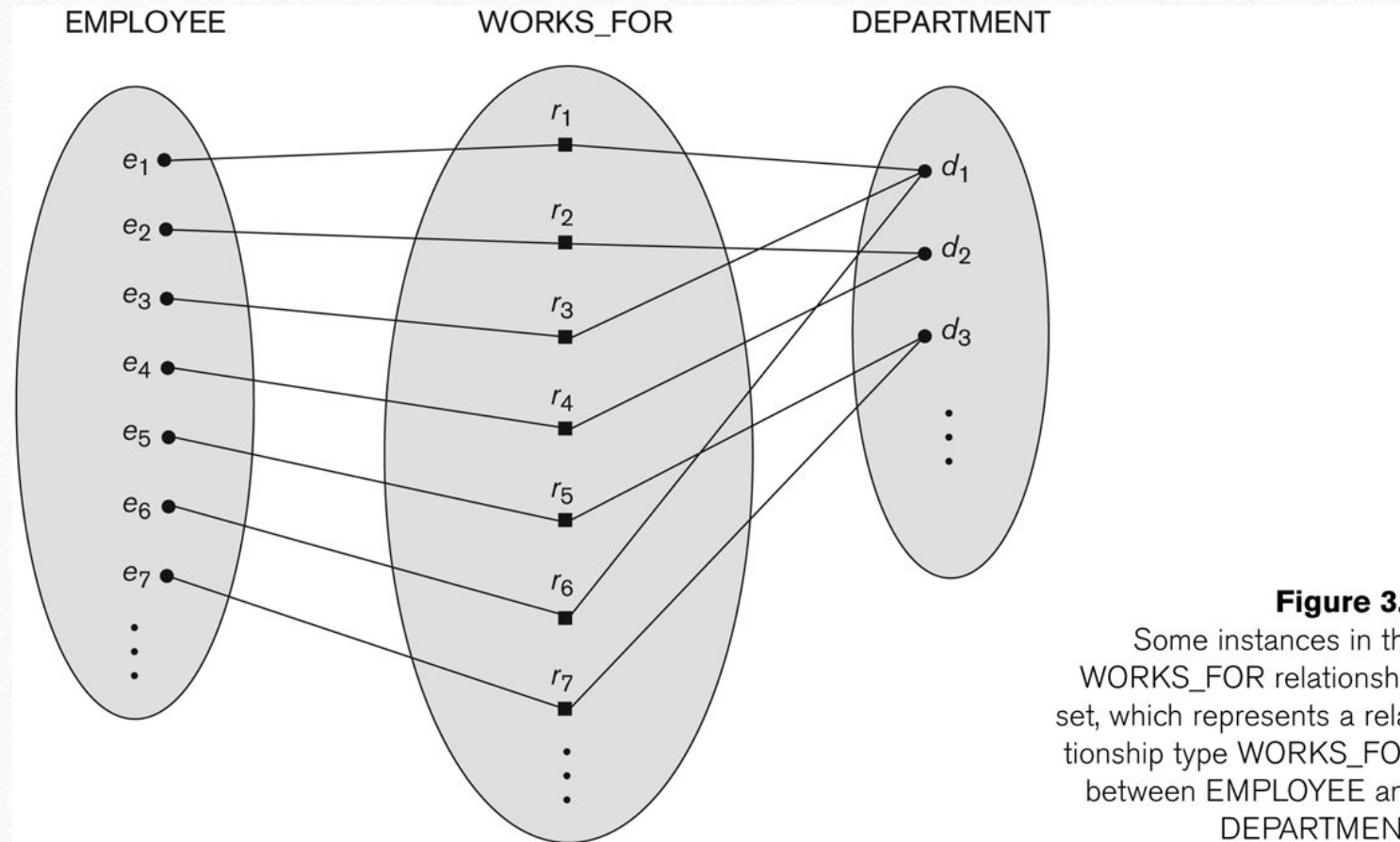
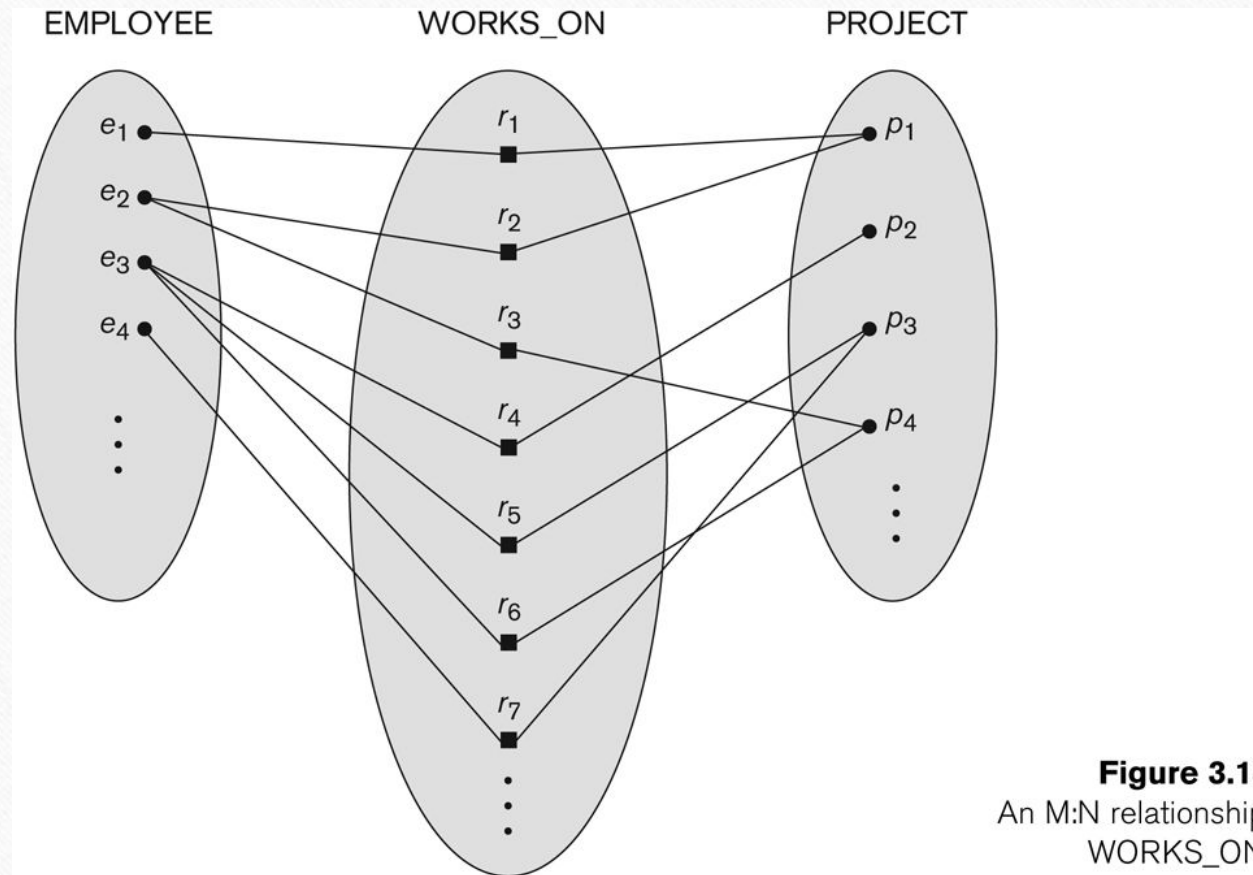


Figure 3.9

Some instances in the WORKS_FOR relationship set, which represents a relationship type WORKS_FOR between EMPLOYEE and DEPARTMENT.

Relationship instances of the M:N WORKS_ON relationship between EMPLOYEE and PROJECT



Relationship type vs. relationship set (1)

- Relationship Type:
 - Is the schema description of a relationship
 - Identifies the relationship name and the participating entity types
 - Also identifies certain relationship constraints
- Relationship Set:
 - The current set of relationship instances represented in the database
 - The current *state* of a relationship type

Relationship type vs. relationship set (2)

- Each instance in the set relates individual participating entities – one from each participating entity type
- In ER diagrams, we represent the *relationship type* as follows:
 - Diamond-shaped box is used to display a relationship type
 - Connected to the participating entity types via straight lines
 - Note that the relationship type is not shown with an arrow. The name should be typically be readable from left to right and top to bottom.

Refining the COMPANY database schema by introducing relationships

- By examining the requirements, six relationship types are identified
- All are *binary* relationships(degree 2)
- Listed below with their participating entity types:
 - WORKS_FOR (between EMPLOYEE, DEPARTMENT)
 - MANAGES (also between EMPLOYEE, DEPARTMENT)
 - CONTROLS (between DEPARTMENT, PROJECT)
 - WORKS_ON (between EMPLOYEE, PROJECT)

ER DIAGRAM – Relationship Types are: WORKS_FOR, MANAGES, WORKS_ON, CONTROLS, SUPERVISION, DEPENDENTS_OF

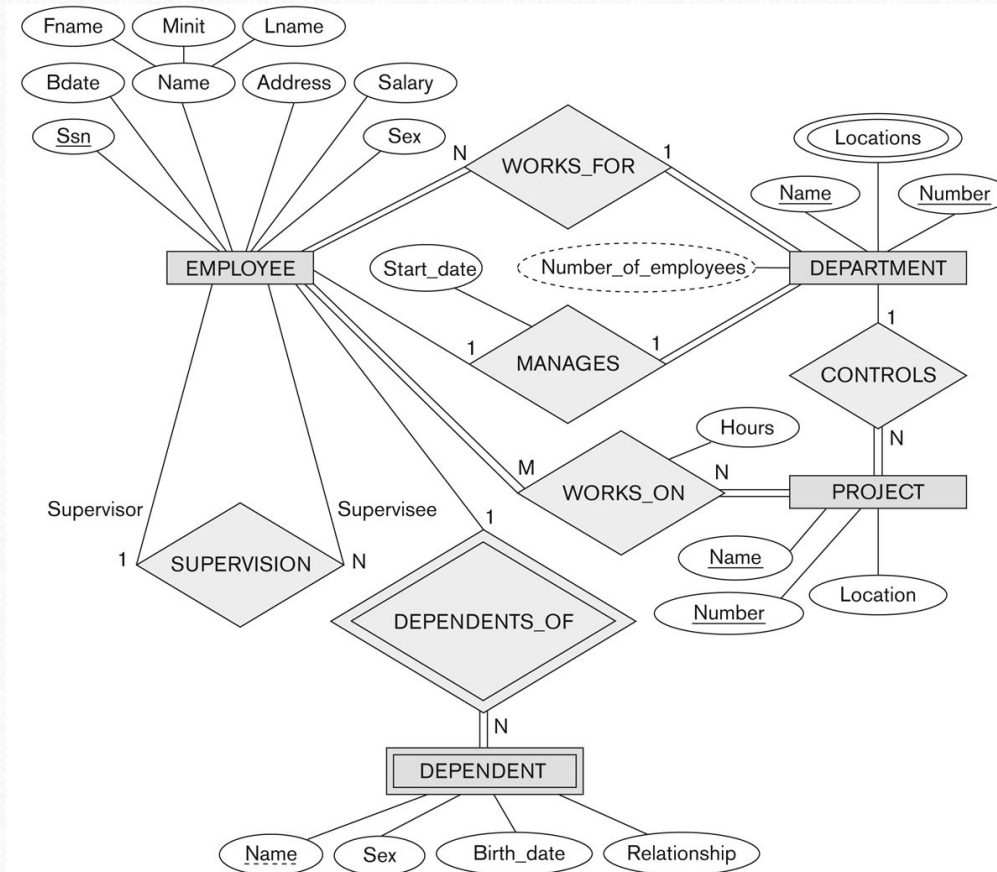


Figure 3.2

An ER schema diagram for the COMPANY database. The diagrammatic notation is introduced gradually throughout this chapter.

Discussion on Relationship Types

- In the refined design, some attributes from the initial entity types are refined into relationships:
 - Manager of DEPARTMENT -> MANAGES
 - Works_on of EMPLOYEE -> WORKS_ON
 - Department of EMPLOYEE -> WORKS_FOR
 - etc
- In general, more than one relationship type can exist between the same participating entity types
 - MANAGES and WORKS_FOR are distinct relationship types between EMPLOYEE and DEPARTMENT
 - Different meanings and different relationship instances.

Constraints on Relationships known as Structural Constraints

- Constraints on Relationship Types
 - (Also known as ratio constraints)
 - Cardinality Ratio (specifies *maximum* participation)
 - One-to-one (1:1)
 - One-to-many (1:N) or Many-to-one (N:1)
 - Many-to-many (M:N)
 - Existence Dependency Constraint (specifies *minimum* participation) (also called participation constraint)
 - zero (optional participation, not existence-dependent, partial) – expressed as single line connecting the participating entity type to the relationship
 - one or more (mandatory participation, existence-dependent, total participation) – expressed as double line connecting the participating entity type to the relationship

Recursive Relationship Type

- A relationship type between the same participating entity type in **distinct roles**
- Also called a **self-referencing** relationship type.
- Example: the SUPERVISION relationship
- EMPLOYEE participates twice in two distinct roles:
 - supervisor (or boss) role
 - supervisee (or subordinate) role
- Each relationship instance relates two distinct EMPLOYEE entities:
 - One employee in *supervisor* role
 - One employee in *supervisee* role

Recursive Relationship Type is: SUPERVISION (participation role names are shown)

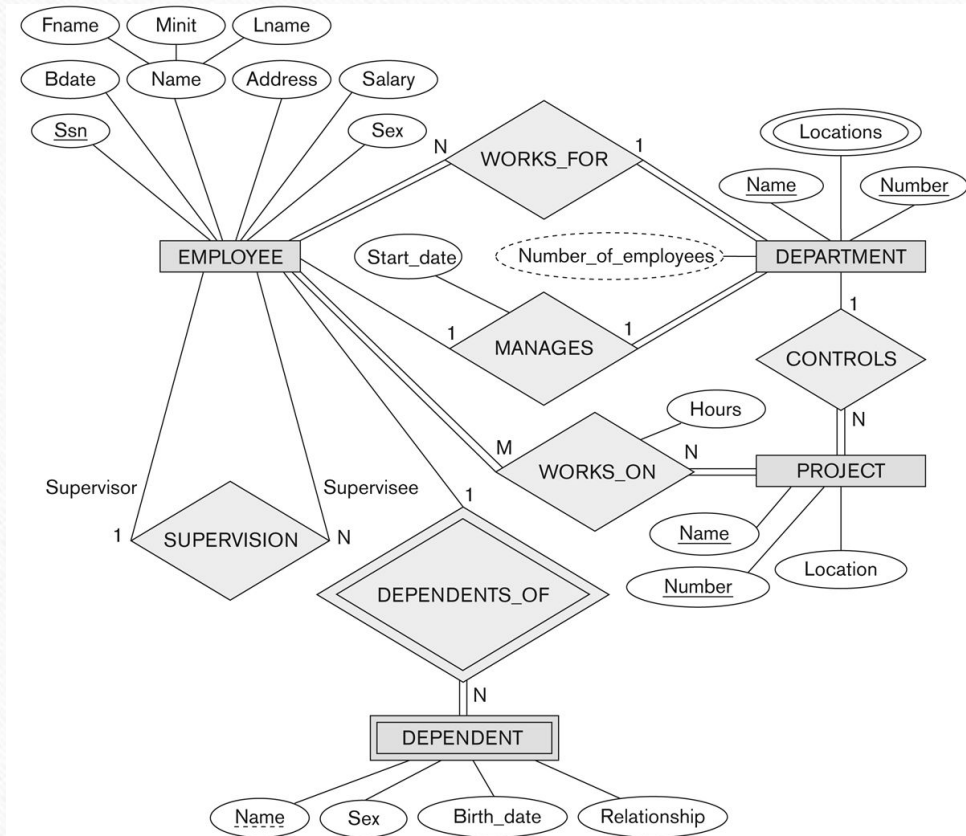


Figure 3.2
An ER schema diagram for the COMPANY database. The diagrammatic notation is introduced gradually throughout this chapter.

Weak Entity Types

- An entity that does not have a key attribute and that is identification-dependent on another entity type.
- A weak entity must participate in an identifying relationship type with an owner or identifying entity type
- Entities are identified by the combination of:
 - A partial key of the weak entity type
 - The particular entity they are related to in the identifying relationship type
- **Example:**
 - A DEPENDENT entity is identified by the dependent's first name, *and* the specific EMPLOYEE with whom the dependent is related
 - Name of DEPENDENT is the *partial key*
 - DEPENDENT is a *weak entity type*
 - EMPLOYEE is its identifying entity type via the identifying relationship type DEPENDENT_OF

Attributes of Relationship types

- A relationship type can have attributes:
 - For example, HoursPerWeek of WORKS_ON
 - Its value for each relationship instance describes the number of hours per week that an EMPLOYEE works on a PROJECT.
 - A value of HoursPerWeek depends on a particular (employee, project) combination
 - Most relationship attributes are used with M:N relationships
 - In 1:N relationships, they can be transferred to the entity type on the N-side of the relationship

Example Attribute of a Relationship Type: Hours of WORKS_ON

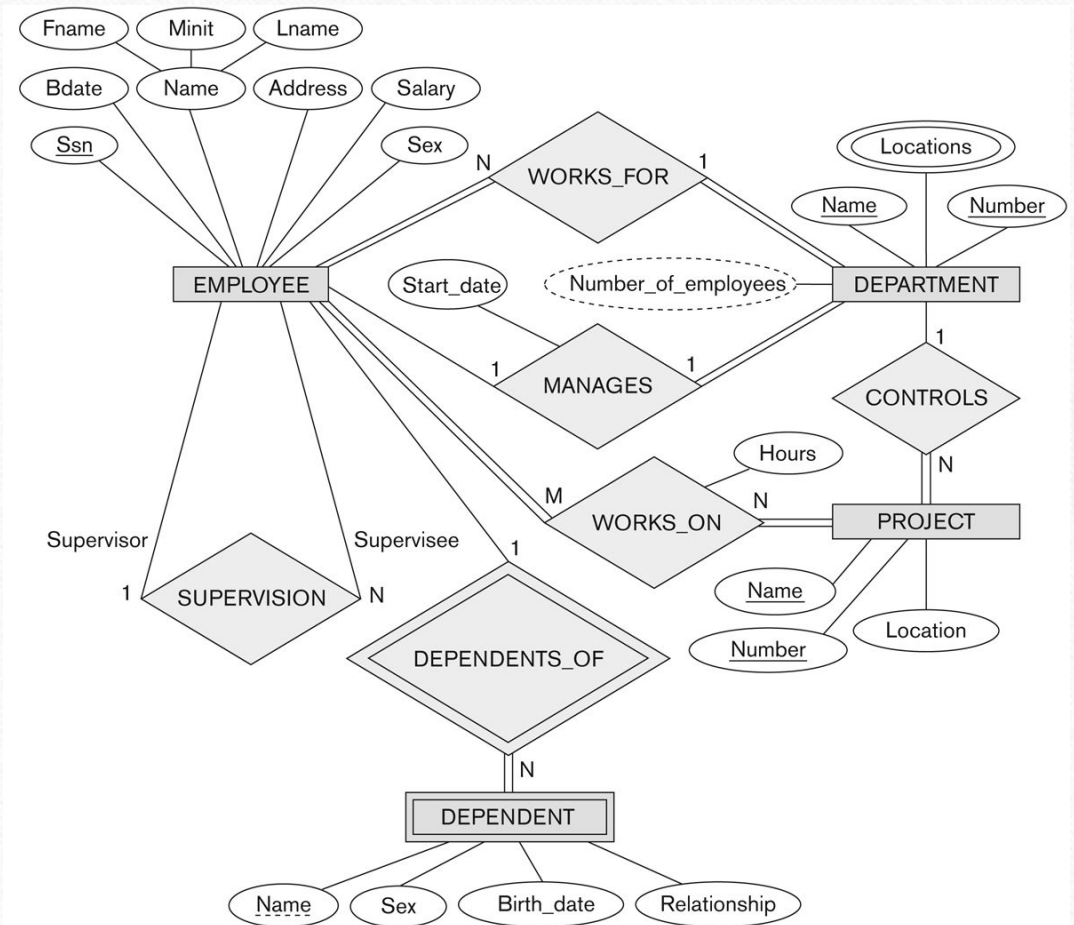


Figure 3.2

An ER schema diagram for the COMPANY database. The diagrammatic notation is introduced gradually throughout this chapter.

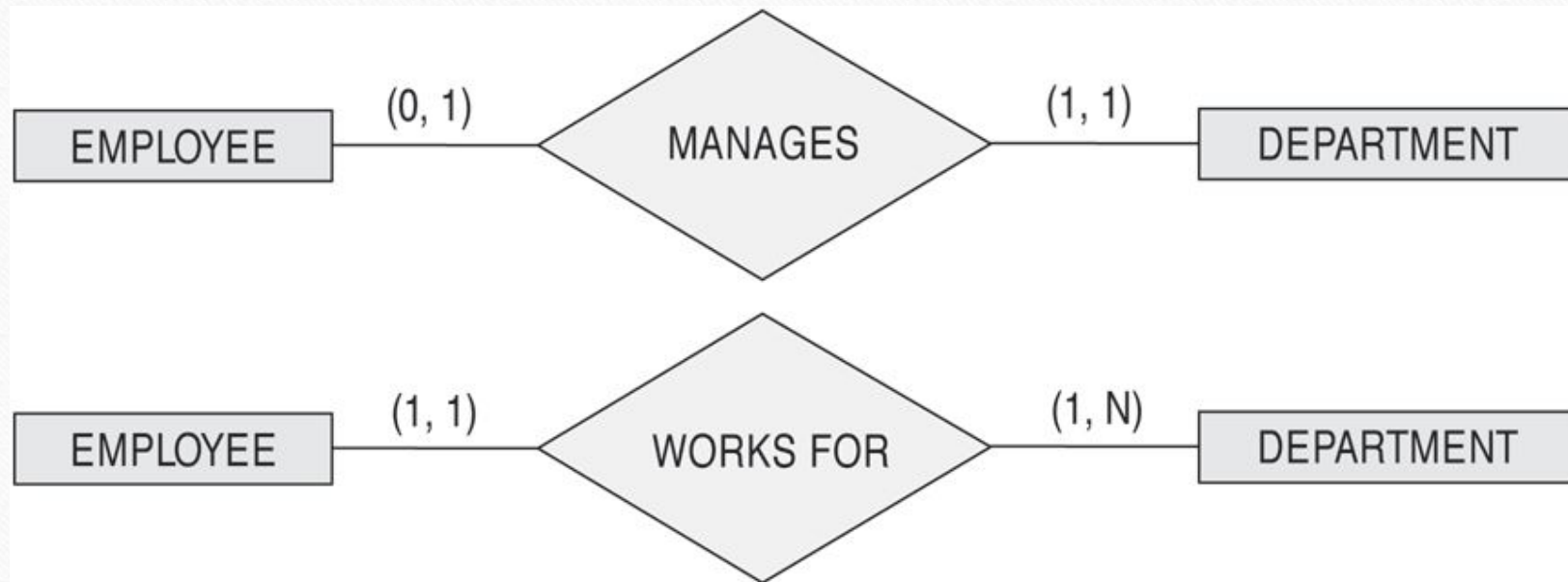
Notation for Constraints on Relationships

- Cardinality ratio (of a binary relationship): 1:1, 1:N, N:1, or M:N
 - Shown by placing appropriate numbers on the relationship edges.
- Participation constraint (on each participating entity type): total (called existence dependency) or partial.
 - Total shown by double line, partial by single line.
- NOTE: These are easy to specify for Binary Relationship Types.

Alternative (min, max) notation for relationship structural constraints:

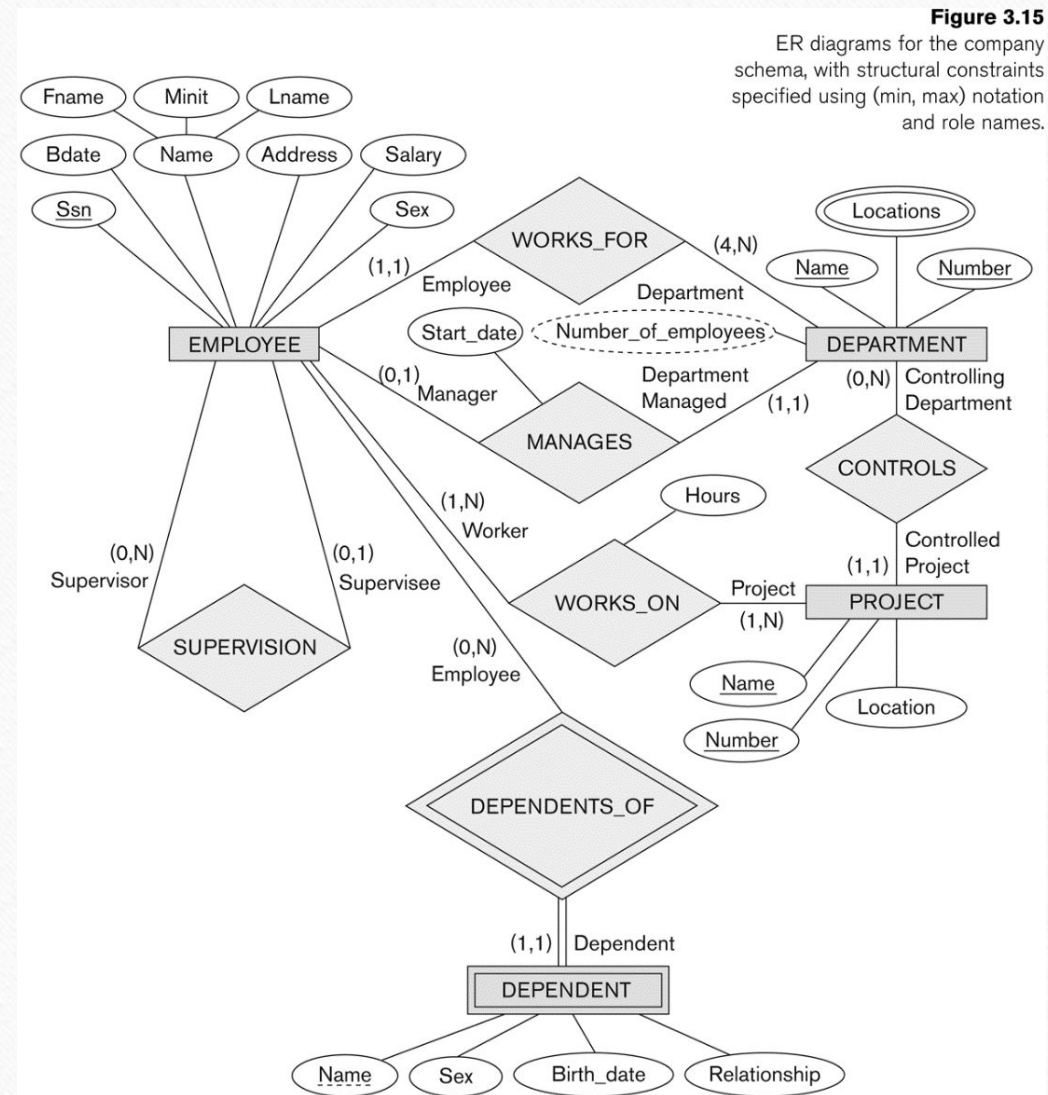
- Specified on each participation of an entity type E in a relationship type R
- Specifies that each entity e in E participates in at least *min* and at most *max* relationship instances in R
- Default(no constraint): min=0, max=n (signifying no limit)
- Must have $\text{min} \leq \text{max}$, $\text{min} \geq 0$, $\text{max} \geq 1$
- Derived from the knowledge of mini-world constraints
- Examples:
 - A department has exactly one manager and an employee can manage at most one department.
 - Specify (0,1) for participation of EMPLOYEE in MANAGES
 - Specify (1,1) for participation of DEPARTMENT in MANAGES
 - An employee can work for exactly one department but a department can have any number of employees.
 - Specify (1,1) for participation of EMPLOYEE in WORKS_FOR
 - Specify (0,n) for participation of DEPARTMENT in WORKS_FOR

The (min,max) notation for relationship constraints



Read the min,max numbers next to the entity type and looking **away from** the entity type

COMPANY ER Schema Diagram using (min, max) notation



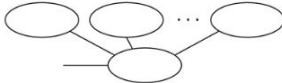
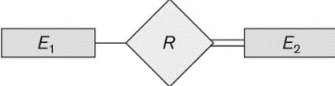
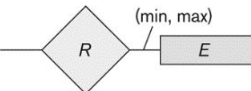
Alternative diagrammatic notation

- ER diagrams is one popular example for displaying database schemas
- Many other notations exist in the literature and in various database design and modeling tools
- Appendix A illustrates some of the alternative notations that have been used
- UML class diagrams is representative of another way of displaying ER concepts that is used in several commercial design tools

Summary of notation for ER diagrams

Figure 3.14

Summary of the notation for ER diagrams.

Symbol	Meaning
	Entity
	Weak Entity
	Relationship
	Identifying Relationship
	Attribute
	Key Attribute
	Multivalued Attribute
	Composite Attribute
	Derived Attribute
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	Cardinality Ratio 1: N for $E_1:E_2$ in R
	Structural Constraint (min, max) on Participation of E in R

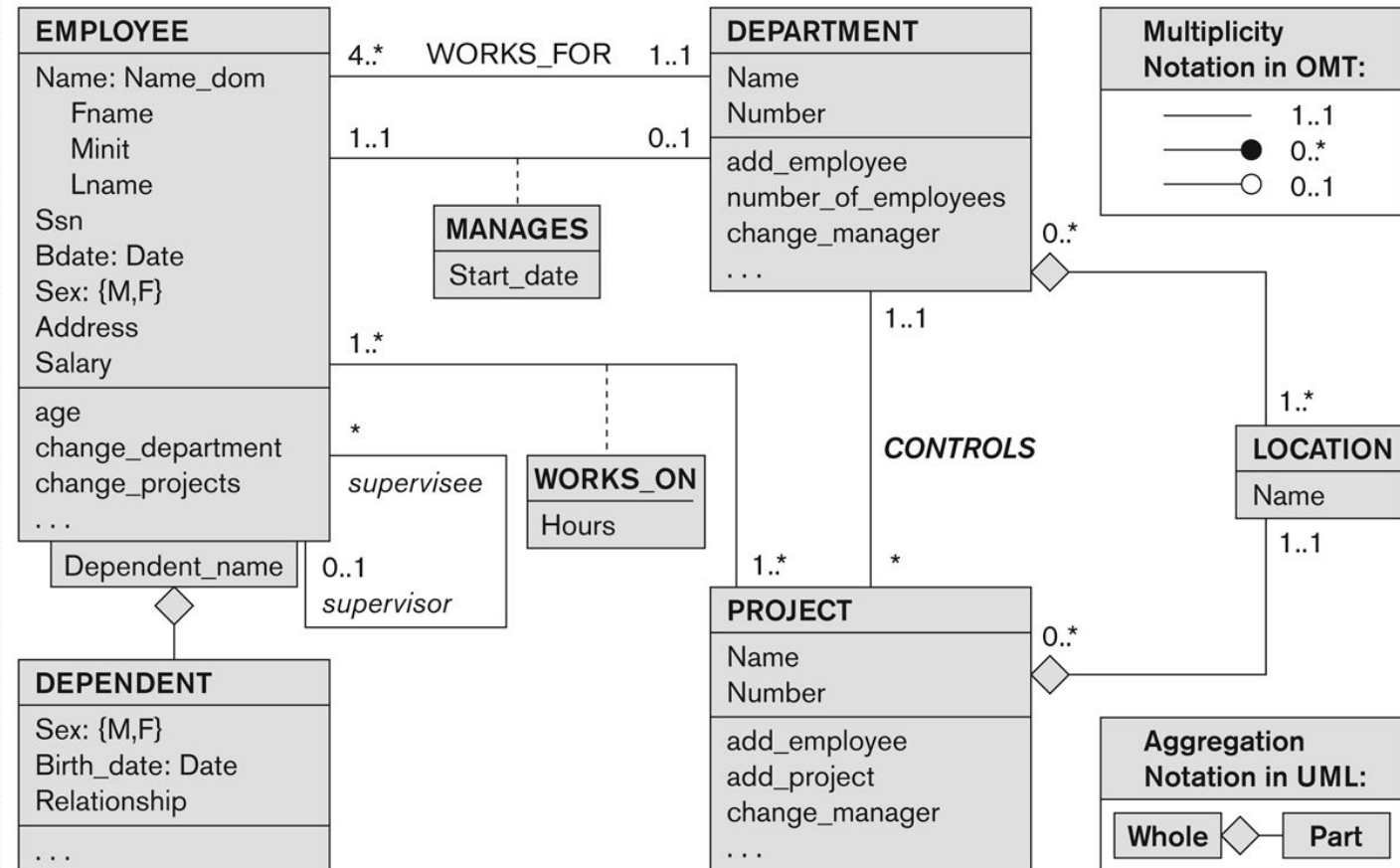
UML class diagrams

- Represent classes (similar to entity types) as large rounded boxes with three sections:
 - Top section includes entity type (class) name
 - Second section includes attributes
 - Third section includes class operations (operations are not in basic ER model)
- Relationships (called associations) represented as lines connecting the classes
 - Other UML terminology also differs from ER terminology
- Used in database design and object-oriented software design
- UML has many other types of diagrams for software design

UML class diagram for COMPANY database schema

Figure 3.16

The COMPANY conceptual schema in UML class diagram notation.



Other alternative diagrammatic notations

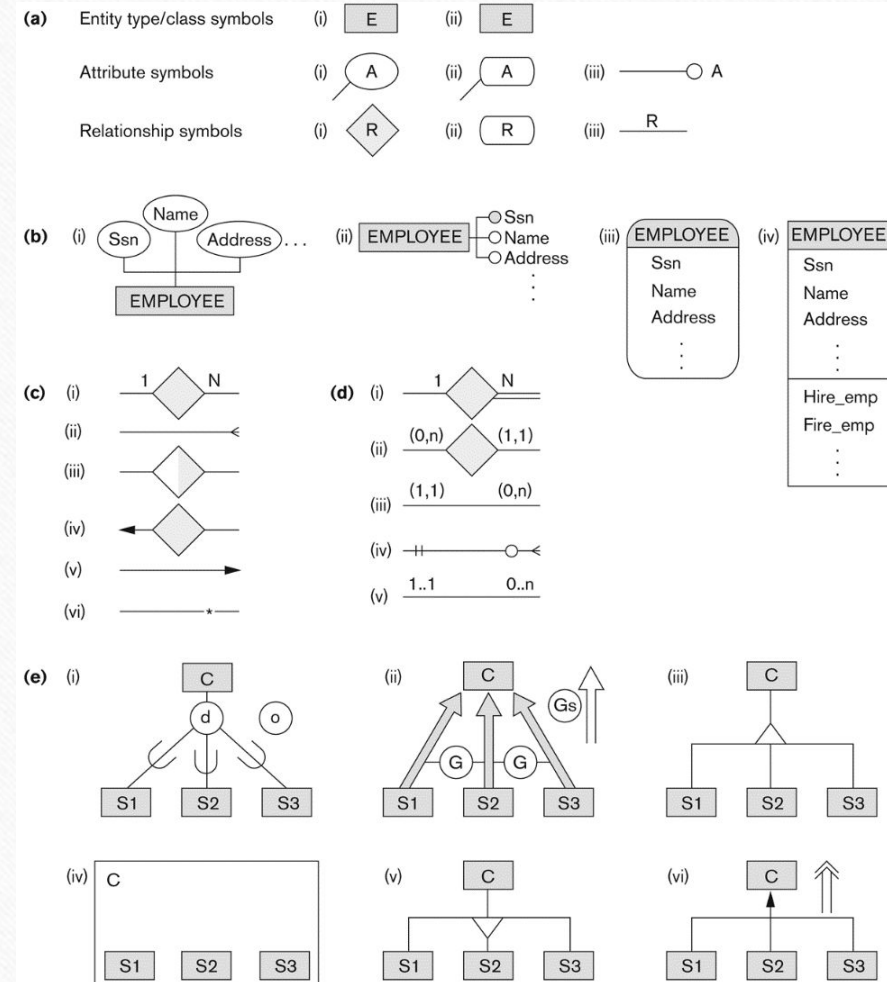


Figure A.1

Alternative notations. (a) Symbols for entity type/class, attribute, and relationship. (b) Displaying attributes. (c) Displaying cardinality ratios. (d) Various (min, max) notations. (e) Notations for displaying specialization/generalization.

Relationships of Higher Degree

- Relationship types of degree 2 are called binary
- Relationship types of degree 3 are called ternary and of degree n are called n -ary
- In general, an n -ary relationship is not equivalent to n binary relationships
- Constraints are harder to specify for higher-degree relationships ($n > 2$) than for binary relationships

Discussion of n-ary relationships ($n > 2$)

- In general, 3 binary relationships can represent different information than a single ternary relationship (see Figure 3.17a and b on next slide)
- If needed, the binary and n-ary relationships can all be included in the schema design (see Figure 3.17a and b, where all relationships convey different meanings)
- In some cases, a ternary relationship can be represented as a weak entity if the data model allows a weak entity type to have multiple identifying relationships (and hence multiple owner entity types) (see Figure 3.17c)

Example of a ternary relationship

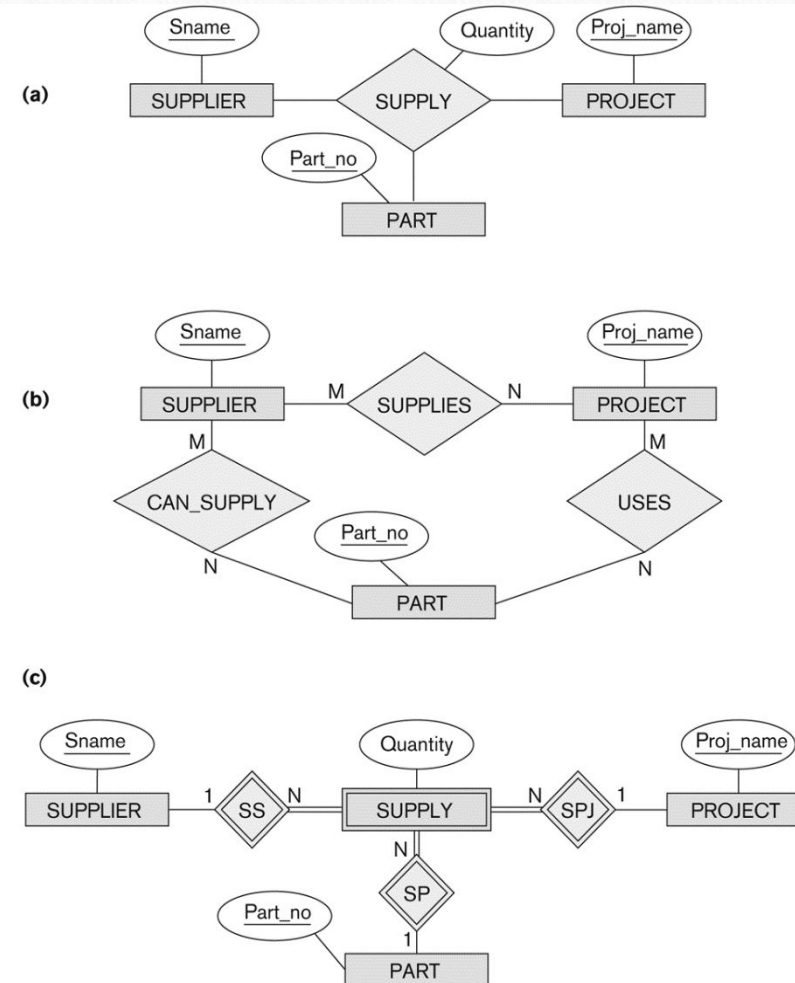


Figure 3.17

Ternary relationship types. (a) The SUPPLY relationship. (b) Three binary relationships not equivalent to SUPPLY. (c) SUPPLY represented as a weak entity type.

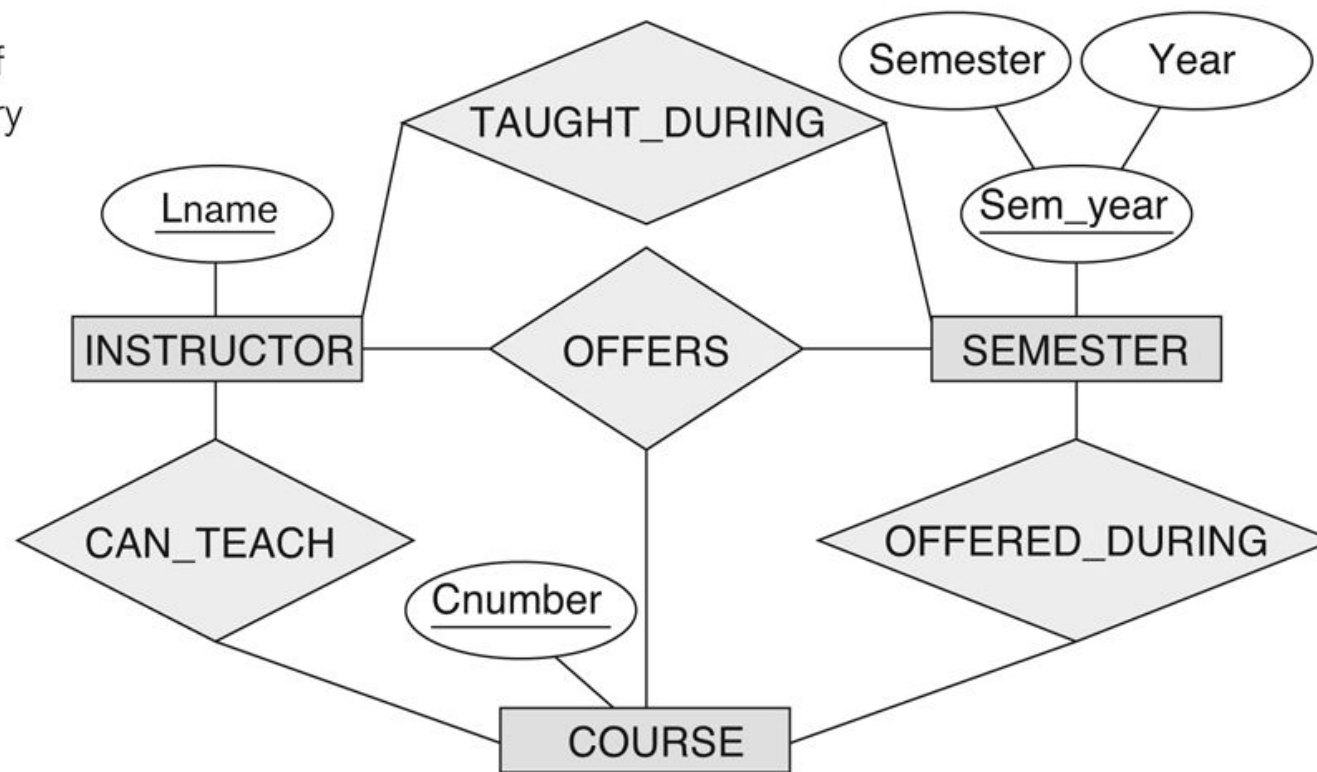
Discussion of n-ary relationships ($n > 2$)

- If a particular binary relationship can be derived from a higher-degree relationship at all times, then it is redundant
- For example, the TAUGHT_DURING binary relationship in Figure 3.18 (see next slide) can be derived from the ternary relationship OFFERS (based on the meaning of the relationships)

Another example of a ternary relationship

Figure 3.18

Another example of ternary versus binary relationship types.



Displaying constraints on higher-degree relationships

- The (min, max) constraints can be displayed on the edges – however, they do not fully describe the constraints
- Displaying a 1, M, or N indicates additional constraints
 - An M or N indicates no constraint
 - A 1 indicates that an entity can participate in at most one relationship instance *that has a particular combination of the other participating entities*
- In general, both (min, max) and 1, M, or N are needed to describe fully the constraints
- Overall, the constraint specification is difficult and possibly ambiguous when we consider relationships of a degree higher than two.

Another Example: A UNIVERSITY Database

- To keep track of the enrollments in classes and student grades, another database is to be designed.
- It keeps track of the COLLEGEs, DEPARTMENTs within each college, the COURSEs offered by departments, and SECTIONs of courses, INSTRUCTORs who teach the sections etc.

UNIVERSITY database conceptual schema

